Database Creation and Table Implementation

MySQL 8.0 DDL for the finalized nine-table design

Prepared by: Vigneshvar K K

Project: Library Management System. **Output convention:** result tables below are reproducible expected outputs from the fixed rows in the shared SQL script. They are not screenshots of MySQL execution.

1. Database creation

This assignment creates the finalized MySQL schema. The same nine-table DDL is used by the shared SQL script and the later reports, so all foreign keys and query examples refer to one design.

```
CREATE DATABASE IF NOT EXISTS library_management_assignment
    CHARACTER SET utf8mb4 COLLATE utf8mb4_0900_ai_ci;
USE library_management_assignment;

-- Rebuild only the assignment tables in reverse dependency order.
DROP TABLE IF EXISTS reservations;
DROP TABLE IF EXISTS loans;
DROP TABLE IF EXISTS book_copies;
DROP TABLE IF EXISTS book_authors;
DROP TABLE IF EXISTS books;
DROP TABLE IF EXISTS members;
DROP TABLE IF EXISTS categories;
DROP TABLE IF EXISTS authors;
DROP TABLE IF EXISTS publishers;
```

2. Table implementation

All tables use InnoDB. Surrogate integer primary keys identify entities; natural identifiers such as ISBN, member number, email, publisher name, and accession code receive unique constraints. Junction and transaction tables use foreign keys to preserve relationships.

```
CREATE TABLE publishers (
    publisher_id INT UNSIGNED NOT NULL,
    publisher_name VARCHAR(120) NOT NULL,
    city VARCHAR(80) NOT NULL,
    PRIMARY KEY (publisher_id),
    CONSTRAINT uq_publishers_name UNIQUE (publisher_name)
) ENGINE = InnoDB;

CREATE TABLE authors (
    author_id INT UNSIGNED NOT NULL,
    author_name VARCHAR(120) NOT NULL,
    country VARCHAR(80) NOT NULL,
    PRIMARY KEY (author_id)
) ENGINE = InnoDB;

CREATE TABLE categories (
    category_id INT UNSIGNED NOT NULL,
    category_name VARCHAR(80) NOT NULL,
    PRIMARY KEY (category_id),
    CONSTRAINT uq_categories_name UNIQUE (category_name)
) ENGINE = InnoDB;

CREATE TABLE books (
    book_id INT UNSIGNED NOT NULL,
    isbn CHAR(13) NOT NULL,
    title VARCHAR(180) NOT NULL,
    publication_year SMALLINT UNSIGNED NOT NULL,
    publisher_id INT UNSIGNED NOT NULL,
    category_id INT UNSIGNED NOT NULL,
    list_price DECIMAL(8, 2) NOT NULL,
    PRIMARY KEY (book_id),
    CONSTRAINT uq_books_isbn UNIQUE (isbn),
```

```
    CONSTRAINT chk_books_year CHECK (publication_year BETWEEN 1450 AND 2100),
    CONSTRAINT chk_books_price CHECK (list_price >= 0),
    CONSTRAINT fk_books_publisher FOREIGN KEY (publisher_id)
        REFERENCES publishers (publisher_id) ON UPDATE CASCADE ON DELETE RESTRICT,
    CONSTRAINT fk_books_category FOREIGN KEY (category_id)
        REFERENCES categories (category_id) ON UPDATE CASCADE ON DELETE RESTRICT
) ENGINE = InnoDB;

CREATE TABLE book_authors (
    book_id INT UNSIGNED NOT NULL,
    author_id INT UNSIGNED NOT NULL,
    author_role VARCHAR(20) NOT NULL DEFAULT 'Author',
    PRIMARY KEY (book_id, author_id),
    CONSTRAINT chk_book_authors_role
        CHECK (author_role IN ('Author', 'Editor', 'Translator')),
    CONSTRAINT fk_book_authors_book FOREIGN KEY (book_id)
        REFERENCES books (book_id) ON UPDATE CASCADE ON DELETE CASCADE,
    CONSTRAINT fk_book_authors_author FOREIGN KEY (author_id)
        REFERENCES authors (author_id) ON UPDATE CASCADE ON DELETE RESTRICT
) ENGINE = InnoDB;

CREATE TABLE book_copies (
    copy_id INT UNSIGNED NOT NULL,
    book_id INT UNSIGNED NOT NULL,
    accession_code VARCHAR(20) NOT NULL,
    shelf_location VARCHAR(30) NOT NULL,
    copy_status VARCHAR(12) NOT NULL DEFAULT 'AVAILABLE',
    PRIMARY KEY (copy_id),
    CONSTRAINT uq_book_copies_accession UNIQUE (accession_code),
    CONSTRAINT chk_book_copies_status
        CHECK (copy_status IN ('AVAILABLE', 'ON_LOAN', 'LOST')),
    CONSTRAINT fk_book_copies_book FOREIGN KEY (book_id)
        REFERENCES books (book_id) ON UPDATE CASCADE ON DELETE RESTRICT
) ENGINE = InnoDB;

CREATE TABLE members (
    member_id INT UNSIGNED NOT NULL,
    member_no VARCHAR(16) NOT NULL,
    full_name VARCHAR(120) NOT NULL,
    email VARCHAR(160) NOT NULL,
    phone VARCHAR(24) NOT NULL,
    joined_on DATE NOT NULL DEFAULT (CURRENT_DATE),
    PRIMARY KEY (member_id),
    CONSTRAINT uq_members_member_no UNIQUE (member_no),
    CONSTRAINT uq_members_email UNIQUE (email)
) ENGINE = InnoDB;

CREATE TABLE loans (
    loan_id INT UNSIGNED NOT NULL,
    copy_id INT UNSIGNED NOT NULL,
    member_id INT UNSIGNED NOT NULL,
    loan_date DATE NOT NULL,
    due_date DATE NOT NULL,
    return_date DATE NULL,
    PRIMARY KEY (loan_id),
    CONSTRAINT chk_loans_dates
        CHECK (due_date >= loan_date AND
            (return_date IS NULL OR return_date >= loan_date)),
    CONSTRAINT fk_loans_copy FOREIGN KEY (copy_id)
        REFERENCES book_copies (copy_id) ON UPDATE CASCADE ON DELETE RESTRICT,
    CONSTRAINT fk_loans_member FOREIGN KEY (member_id)
        REFERENCES members (member_id) ON UPDATE CASCADE ON DELETE RESTRICT
) ENGINE = InnoDB;

CREATE TABLE reservations (
    reservation_id INT UNSIGNED NOT NULL,
    member_id INT UNSIGNED NOT NULL,
    book_id INT UNSIGNED NOT NULL,
    reservation_date DATE NOT NULL,
    reservation_status VARCHAR(12) NOT NULL DEFAULT 'PENDING',
    PRIMARY KEY (reservation_id),
    CONSTRAINT chk_reservations_status
        CHECK (reservation_status IN ('PENDING', 'FULFILLED', 'CANCELLED')),
    CONSTRAINT fk_reservations_member FOREIGN KEY (member_id)
        REFERENCES members (member_id) ON UPDATE CASCADE ON DELETE RESTRICT,
    CONSTRAINT fk_reservations_book FOREIGN KEY (book_id)
        REFERENCES books (book_id) ON UPDATE CASCADE ON DELETE RESTRICT
) ENGINE = InnoDB;
```

3. Table map

Table	Purpose	Main relationship
publishers	Publisher name and city	One publisher can publish many books.
categories	Controlled book subject groups	One category can describe many books.
authors	Author identity and country	Connected to books through book_authors.
books	Catalog edition/title and price	References one publisher and category.
book_authors	Book-to-author association and role	Composite key prevents duplicate links.
book_copies	Physical copy and shelf/status	Each copy belongs to one book.
members	Library membership and contact	Member identity is unique.
loans	Checkout and return dates	References one copy and one member.
reservations	Member request for a book	References one member and one book.

4. Expected schema check

```
USE library_management_assignment;
SHOW TABLES;
```

Expected output from the deterministic sample rows in Library_Management_System_MySQL_8.sql:

Expected table names
authors
book_authors
book_copies
books
categories
loans
members
publishers
reservations

Run and capture

Run the full DDL from the shared file or this report in MySQL Workbench. Keep the schema name visible and capture the SQL plus the successful table list. The DDL uses named CHECK constraints; use MySQL 8.0.16 or later for their enforcement.

The script drops and recreates the nine tables in `library_management_assignment` to make a clean assignment run. Use that dedicated schema and back up any rows there before rerunning it.